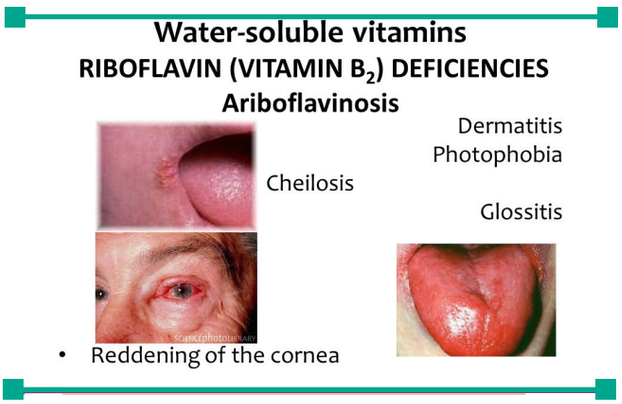
and death occurs due to heart failure. Infantile beri beri is seen in many South East Asian countries where the diets consist mostly of "polished rice".

(b) Riboflavin (B2): Riboflavin is a stable vitamin which is resistant to acid, heat and oxidation. But it is unstable in the presence of alkali and light.

- **Functions:**
  The two coenzymes of Riboflavin Flavin Mono Nucleotide (FMN) and Flavin Adenine Dinucleotide (FAD) perform the following functions
  - Formation of red blood cells in the bone marrow.
  - Regulates the functions of hormones in carbohydrate metabolism.
  - Present in the retina in the free form which gets converted to a compound which stimulates the optic nerve.
  - Release of energy from glucose, amino acids and fatty acids.
- **Food sources:** Good sources of riboflavin are milk and milk products, eggs, liver, whole or enriched grains and green leafy vegetables.



- **Deficiency:** The deficiency of riboflavin (ariboflavinosis) leads to glossitis (swollen and reddened tongue), swollen lips, cheilosis (inflammation of the corners of the mouth, are some of the common symptoms observed. Further deficiency states are marked by chronic conditions like tuberculosis, prolonged fevers, malabsorption, hyperthyroidism and malignancy.



(c) Niacin: Niacin formerly known as nicotinic acid was obtained by the oxidation of nicotinic acid. Apart from the food sources, Niacin is also obtained from tryptophan (60mg) an essential amino acid which can be converted into niacin (1mg).

- **Functions:** Two coenzymes of Niacin Nicotinamide Adenine Dinucleotide (NAD) Nicotinamide Adenine Dinucleotide Phosphate (NADP) are required for:
  - Release of energy from all energy yielding nutrients like carbohydrate, protein and fat.
  - Normal functioning of the skin, intestinal tract and the nervous system.

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